PROJECT MANAGER

Software Requirements
Specifications for
Project Manager

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Prepared By

Ethan Van Drunen

Diego Chavez Espinosa

ECE 35400 Software Engineering Design I

Purdue University Northwest

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Document Approvals

Name	Role	Signature
Ethan Van Drunen	Developer	
Diego Chavez Espinosa	Developer	

Project Information

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1. Introduction

1.1 Purpose of the document

This document lists every requirement for Project Manager, the web application our team is building for the ECE 35400 final project. It covers what the system must do (functional requirements), how well it must do it (non-functional requirements), and the UML design that meets them.

It brings together our Phase 1 proposal, the Phase 2.1 user stories and class diagram, the Phase 2.2 sequence diagrams, and the instructor's feedback on each. It is the baseline the team will build and test against.

1.2 Scope of the project

Project Manager is a website where managers create projects, give out tasks and set due dates, and team members update the status of their own tasks. Progress, late tasks and workloads are worked out from those updates, so nobody has to ask for a status report.

The software system will include the following key features:

• **Project and task management:** Managers create projects and tasks, assign tasks to one or more team members, and set and change due dates.

• **Task status updates:** Team members mark their tasks as Not Started, In Progress or Completed.

• **Progress tracking:** The system shows the percent done for each project and each person's share of the finished work.

• **Late-task alerts and workload:** Late tasks are flagged with an in-app alert, and each person's open-task count is shown.

• **Login and access control:** Managers and team members log in, and only managers can use manager features.

The scope of this project does not include a time clock or payroll, file sharing, chat or video calls, advanced scheduling charts (such as Gantt charts), email or text alerts (alerts appear only in the app), a phone app, or connections to other apps such as Slack or Google Calendar.

1.3 Stakeholders involved

A user works with the software directly. A stakeholder is affected by it or depends on it.

Role or group	Type	Main need or concern
Managers / project leads	Primary user	See how every project and person is doing without check-ins; create projects, give out tasks, set due dates.
Team members / employees	Primary user	A clear list of their own tasks and due dates; progress reaches the manager without writing reports.

System administrator	Secondary user	Set up and remove accounts; decide who is a manager and who is a team member.
Company leadership	Stakeholder	Projects finished on time and fair workloads.
Clients	Stakeholder	Can count on the due dates the system helps managers meet.
Development team (Ethan and Diego)	Stakeholder	A project small enough for two people to finish this semester.

1.4 Overview of Project Manager

Today most teams track work by hand. Tasks are handed out in meetings, emails and texts, and progress lives in spreadsheets. A manager only knows how a project is going by checking in with each person. This wastes time, hides late work until it is already late, and lets workloads get uneven.

Project Manager keeps tasks, due dates and progress in one place. It will consist of the following main features:

1.4.1 Projects

Each project has a name, description, start date and due date, plus a progress percent worked out from its tasks.

1.4.2 Tasks

Tasks belong to one project, are assigned to one or more team members, and carry a status and a due date.

1.4.3 Manager Dashboard

Every project appears on one screen with its percent done, late-task count and next due date.

1.4.4 Late Alerts and Workload

Late tasks are flagged in the app, and each person's open-task count is shown so work can be split fairly.

2. Overall Description

2.1 Product perspective

Project Manager is a standalone web application. It runs in a normal web browser on a computer or phone and does not connect to any outside system. All alerts are shown inside the app.

It has two user interfaces, chosen at login by the user's role:

2.1.1 Manager Interface

Managers create and edit projects, create and assign tasks, set due dates, and view the dashboard, late tasks and workloads.

2.1.2 Team Member Interface

Team members view their own tasks and due dates, update task status, and see late-task alerts.

2.2 Product functions

Project Manager will provide the following functions:

Function	Who uses it	Requirements
Create and edit projects	Manager	FR-1, FR-3
Create tasks and assign them	Manager	FR-2
Set and change due dates	Manager	FR-3
Update task status	Team member	FR-4
Track project progress and each person's share	Manager	FR-5
Dashboard of all projects	Manager	FR-6
Late-task flags and alerts	Manager, team member	FR-7
Workload per person	Manager	FR-8
Log in and role-based access	All users	FR-9

2.3 User characteristics

• **Managers** run one or more projects and teams. They may have little technical background, so creating a project and assigning a task must take no training (NFR-4).

• **Team members** mostly read their task list and change task status. They need a short, clear task list, not reporting tools.

• **System administrator** sets up accounts and roles. This is the only user who needs to know how the system is configured.

• **Language:** The system will be available in English only.

2.4 Constraints

• **Platform:** The system must work in any normal web browser on a computer or phone; there is no native phone app.

• **Team size:** The system must work for both small teams and large companies, so nothing can be built around one team size.

• **Access:** Everyone must log in. Only managers can create projects, give out tasks or change due dates.

• **Ethics and privacy:** Workload and progress data is for setting fair due dates and workloads, not for punishing people. A person's workload is visible only to them and their manager.

• **Schedule and team:** Two developers, finished within the Fall 2026 semester.

2.5 Assumptions and dependencies

Assumptions:

- Each user has one account with one role, Manager or Team member.
- Accounts and roles are set up by the system administrator before a user's first login.
- Users have a modern browser and an internet connection.
- The server clock is correct, since late tasks (FR-7) depend on comparing the current date to each due date.

Dependencies:

- A web server and database to store users, projects and tasks.
- A password hashing library for NFR-3.
- A daily backup job for NFR-5.

2.6 Apportioning of requirements

Requirements are built in two releases so each one can be tested before the next starts.

Release 1: Core project and task tracking

- FR-1, FR-2, FR-3, FR-4, FR-9
- NFR-3

Release 2: Progress, alerts and workload

- FR-5, FR-6, FR-7, FR-8
- NFR-1, NFR-2, NFR-4, NFR-5, NFR-6, NFR-7

3. Specific Requirements

3.1 Functional Requirements

Each requirement has an ID used throughout this document and a test that shows it works.

3.1.1 Project management

ID	Requirement	User	How we will check it
FR-1	The system shall let a manager create a project with a name, description, start date and due date.	Manager	Create a project and check that all four details are saved and show up in the project list.
FR-3	The system shall let a manager set and change the due date for each task and project.	Manager	Set a due date, change it, and check that the new date shows for both the manager and the team member.

3.1.2 Task assignment and status

ID	Requirement	User	How we will check it
FR-2	The system shall let a manager create a task in a project and assign it to one or more team members.	Manager	Assign a task to two people and check that it shows up on both of their task lists.
FR-4	The system shall let a team member mark their own task as Not Started, In Progress or Completed.	Team member	Change a task to each of the three statuses and check that the manager sees each change.
FR-7	The system shall mark a task as late if it is not Completed by its due date and show an alert to the team member and manager.	Manager, team member	Make a task that was due yesterday and is not finished; check that it is marked late and both people get an alert.

3.1.3 Progress and workload tracking

ID	Requirement	User	How we will check it
FR-5	The system shall show what percent of each project is done and what percent of the finished work each person did.	Manager	Finish 1 of 4 tasks and check that the project shows 25% done.
FR-6	The system shall show the manager a main screen listing all their projects, how much of each is done, how many tasks are late, and the next due date.	Manager	Set up 3 projects with known numbers and check that the main screen shows the right numbers.
FR-8	The system shall show how many unfinished tasks each team member has.	Manager	Give one person 3 tasks and another 1; check that it shows 3 and 1.

3.1.4 Login and access

ID	Requirement	User	How we will check it
FR-9	The system shall require a username and password to log in, and only give managers access to manager features.	All users	Try to open the app without logging in (blocked); log in as a team member and check that manager features are hidden.

3.2 Non-functional Requirements

Non-functional requirements describe how well the system must work, rather than what it does. Each one has a number or a clear pass/fail goal so it can be tested.

3.2.1 Performance requirements

ID	Requirement	How we will check it
NFR-1	The manager's main screen shall load in 2 seconds or less with 50 projects and 1,000 tasks in the system.	Load 50 projects and 1,000 tasks, open the main screen 20 times and time it. At least 19 of 20 loads take 2 seconds or less.
NFR-2	A team member's task update shall show on the manager's screen within 3 seconds.	Update a task as a team member and time how long it takes to appear for the manager.
NFR-6	At least 200 people shall be able to use the system at once without the main screen taking longer than 2 seconds to load.	Use a load-testing tool to simulate 200 users and time the main screen.

3.2.2 Usability requirements

ID	Requirement	How we will check it
NFR-4	A new manager shall be able to create a project and assign a task in under 5 minutes without training.	Have 5 first-time users try it. At least 4 finish in under 5 minutes.

3.2.3 Security and privacy requirements

ID	Requirement	How we will check it
NFR-3	Passwords shall be hashed before they are saved, and an account shall lock for 15 minutes after 5 wrong password tries.	Check the stored data and confirm no password can be read. Enter a wrong password 5 times and confirm the account locks.
NFR-7	Team members shall only see projects they are part of. Only the person and their manager can see that person's workload.	Log in as someone on Project A and try to open Project B or another person's workload. Both are blocked.

3.2.4 Availability and reliability requirements

ID	Requirement	How we will check it
NFR-5	The system shall be up at least 99% of the time and back up all data once a day.	Track downtime over one month. Delete test data and confirm it can be restored from the backup.

3.2.5 Compatibility requirements

a. **Browser Compatibility:** The system shall run in current versions of Google Chrome, Mozilla Firefox, Apple Safari and Microsoft Edge.

b. **Device Compatibility:** The system shall work on desktop and mobile browsers with no install needed (see 2.4).

4. Use Cases

4.1 User stories

#	Source	User story	Acceptance criterion
1	FR-1	As a manager, I want to create a project so that I can keep my projects and their details organized.	Given I want to create a project, when I select Create Project, then I am asked to fill in the project details.
2	FR-4	As a team member, I want to update the status of my tasks so that I can stay on track.	Given a task assigned to me, when I change its status to Completed, then my manager sees the task as Completed.
3	FR-9	As a team member, I want to log in with my username and password so that only I can update my tasks.	Given I have a team member account, when I enter my correct username and password, then I see my task list.
4	NFR-1	As a manager, I want my main screen to show everyone's progress quickly so that I know how each project is going.	Given 50 projects and 1,000 tasks in the system, when I open the main screen, then it is fully displayed within 2 seconds.

4.2 Create a project use case (UC-1, FR-1)

Description: This use case describes how a manager creates a new project with its basic details.

Actors:

- Manager: a user with a Manager account.
- System: Project Manager, which validates and saves the project.

Pre-conditions:

- The manager is logged in with a Manager account.

Post-conditions:

- The new project is saved and appears in the manager's project list with 0% progress.

Basic Flow:

1. The manager selects Create Project.
2. The system asks for name, description, start date and due date.
3. The manager enters the details and submits.
4. The system validates the details.
5. The system saves the project and shows it in the project list.

Alternative Flows:

4a. If a field is empty or the due date is before the start date, the system shows an "invalid details" message and no project is created. The manager can correct the details and submit again.

Exceptions:

If a team member tries to reach Create Project, the option is hidden and access is blocked (FR-9).

4.3 Update task status use case (UC-2, FR-4)

Description: This use case describes how a team member changes the status of a task assigned to them.

Actors:

- Team member: a user assigned to the task.
- System: Project Manager, which checks the assignment, saves the status and updates progress.

Pre-conditions:

- The team member is logged in.
- The task exists in a project the team member is part of.

Post-conditions:

- The task shows the new status to the team member and manager.
- The project's progress percent is recalculated (FR-5).

Basic Flow:

1. The team member opens their task list and selects a task.
2. The team member picks a new status: Not Started, In Progress or Completed.
3. The system checks that the task is assigned to this team member.
4. The system saves the new status and refreshes the project's progress.
5. The manager sees the change within 3 seconds (NFR-2).

Alternative Flows:

3a. If the task is not assigned to this team member, the system blocks the change and shows a "not assigned to this task" message.

Exceptions:

If the system cannot save the change, the task keeps its old status and an error message is shown.

4.4 Log in use case (UC-3, FR-9)

Description: This use case describes how a user logs in and sees the features for their role.

Actors:

- Manager or team member: any user with an account.
- System: Project Manager, which checks the login and the user's role.

Pre-conditions:

- The user has an account set up by the system administrator.

Post-conditions:

- A manager sees the main screen (FR-6). A team member sees their task list.

Basic Flow:

1. The user enters a username and password.
2. The system checks them against the stored, hashed password (NFR-3).
3. The system checks the user's role and shows the matching screen.

Alternative Flows:

2a. If the details are wrong, the system shows an error and counts the failed try.

2b. After 5 wrong tries, the account locks for 15 minutes (NFR-3).

Exceptions:

If a user opens any page without logging in, they are sent to the login page.

5. UML Design

The class diagram and both sequence diagrams use the same class and operation names, so every message in a sequence diagram is an operation in the class diagram.

5.1 Class diagram

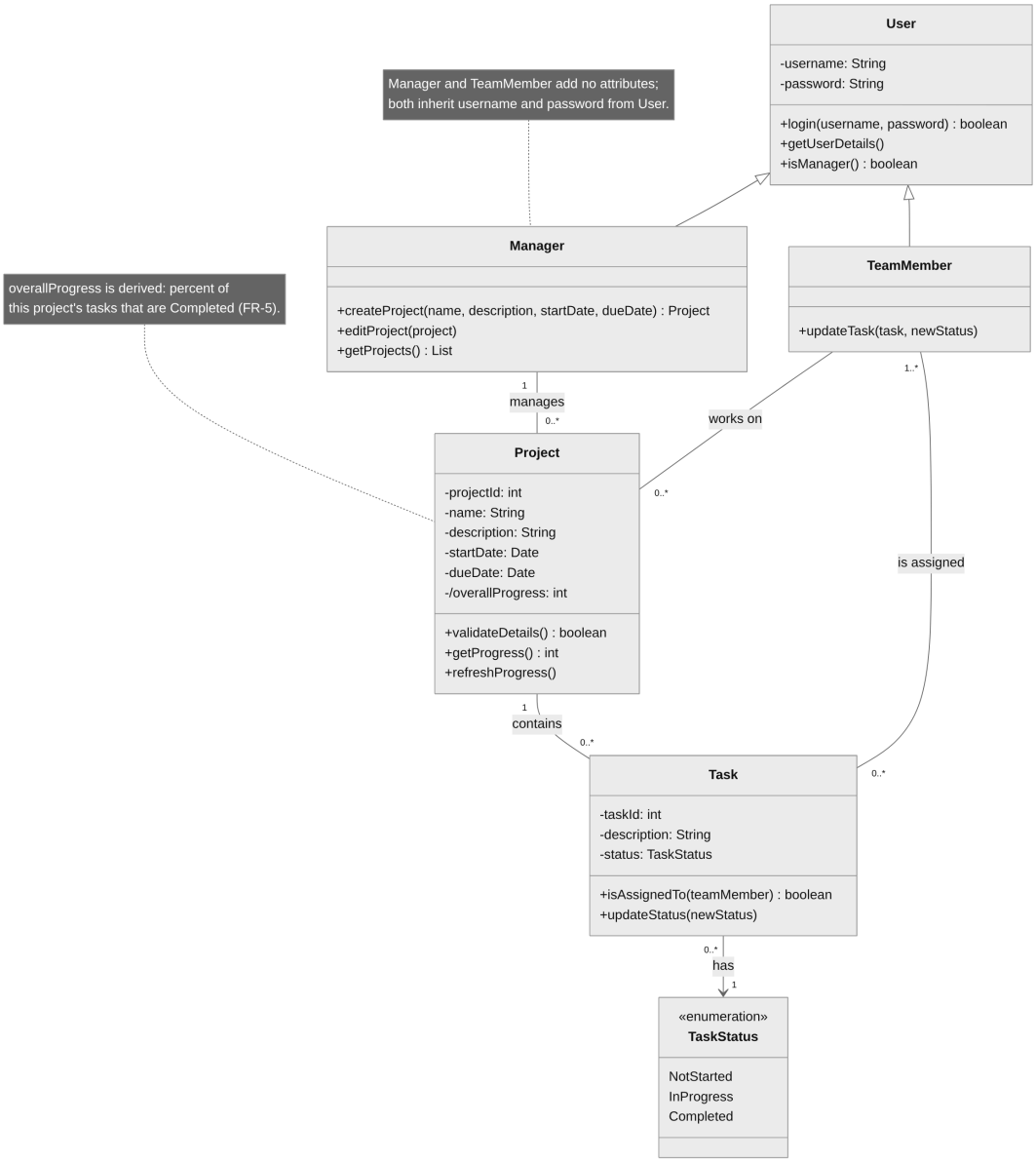


Figure 1. Class diagram for Project Manager

Requirements covered: FR-1 (Manager.createProject, Project), FR-2 (Task assigned to TeamMember), FR-4 (Task.updateStatus, TaskStatus), FR-5 (/overallProgress), FR-9 (User.login, User.isManager).

5.2 Sequence diagram: create a project (FR-1, UC-1)

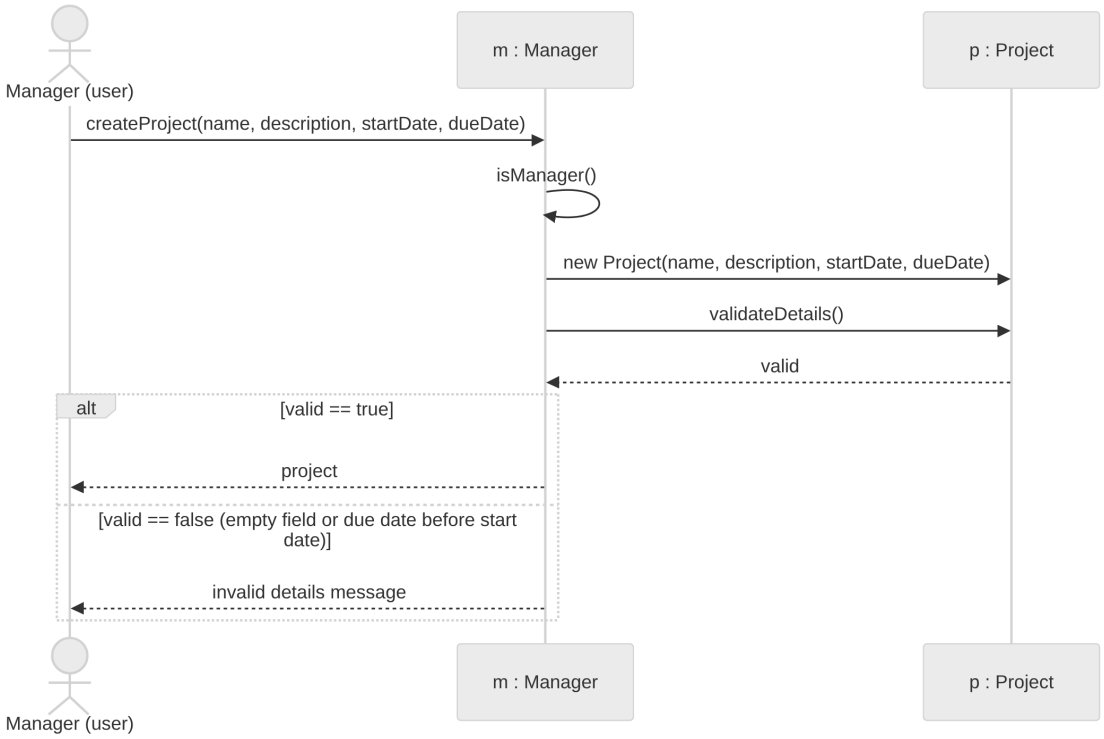


Figure 2. Sequence diagram for creating a project

5.3 Sequence diagram: update task status (FR-4, UC-2)

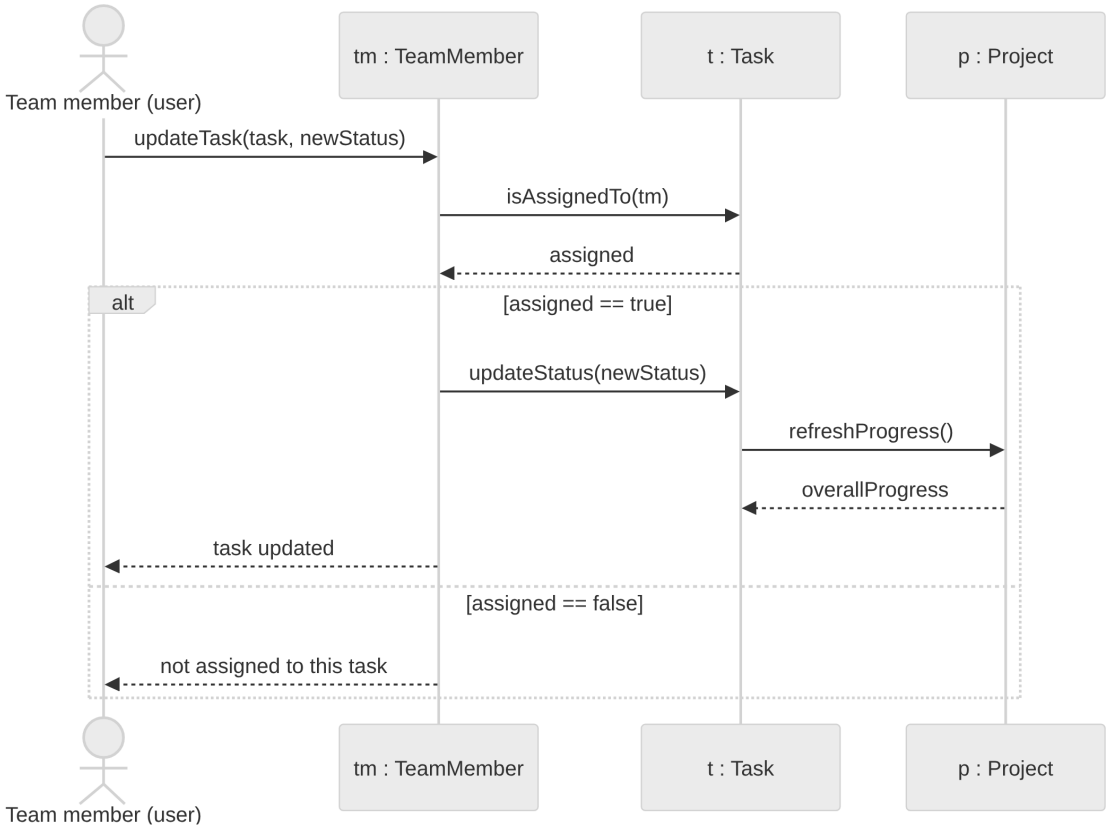


Figure 3. Sequence diagram for updating a task's status

6. Appendices

6.1 Glossary of terms

- Project: A body of work with a name, description, start date and due date, made up of tasks.
- Task: One unit of work in a project, assigned to one or more team members.
- Task status: Not Started, In Progress or Completed.
- Late task: A task not Completed by its due date.
- Workload: The number of unfinished tasks assigned to one person.
- Derived attribute: A value worked out from other data instead of stored on its own, marked with / in UML.
- Use case: A description of one interaction between a user and the system.
- Class diagram: A visual representation of the classes in a system and how they relate.
- Sequence diagram: A visual representation of the messages passed between objects in one scenario, in time order.

6.2 List of acronyms and abbreviations

- FR: Functional Requirement
- NFR: Non-Functional Requirement
- SRS: Software Requirements Specification
- UC: Use Case
- UML: Unified Modeling Language

6.3 Feedback addressed

Phase	Feedback	Where it is fixed
2.1	Story roles mixed up (story 3, story 4)	4.1: story 3 is a team member, story 4 a manager
2.1	Story 4 criterion did not test NFR-1	4.1: criterion now tests the 2-second load only
2.1	Story 2 criterion not checkable	4.1: "my manager sees the task as Completed"
2.1	Story 3 had more than one benefit	4.1: one benefit, "only I can update my tasks"
2.1	Multiplicities missing	5.1: on both ends of every association
2.1	NewProject class should be part of Project	5.1: removed; Manager.createProject() creates a Project
2.1	Task not linked to TeamMember	5.1: TeamMember 1..* is assigned 0..* Task
2.1	Manager and TeamMember had no attributes	5.1: note says both inherit username and password from User
2.1	No login operation	5.1: User.login(username, password): boolean
2.1	Completion percentage on Task	5.1: replaced by TaskStatus enumeration; Project./overallProgress is derived
2.2	Write more organized	5.1 to 5.3: all diagrams redrawn in one style with matching names; typos fixed (updateTask, isAssignedTo)
2.2	Diagrams should match each other	5.2: isManager() and validateDetails() are class-diagram operations; return arrows carry values
Review	"Almost every load" not testable (NFR-1)	3.2.1: at least 19 of 20 loads in 2 seconds or less

6.4 References and resources

- ECE 35400 Case Study, File 4: Sample SRS Document (Y-Mart)
- Project repository: https://github.com/evandrun/projectManager